EE 105 Fall 2025

Homework 3 – upload to grade scope

(Due October 2, before class, late submission will incur 20% points/day penalty)

Instructions: Perform the following tasks based on the circuits and concepts discussed in class. Be sure to show all work where applicable.

Problem 1: What are the different types of materials used in electronics? What are their charge carrier densities?

Problem 2: Why is silicon used in electronics?

Problem 3: Draw the atomic models of the p-type and n-type semiconductors.

Problem 4:

- What is this electronic device?
- Draw the I-V characteristics and explain what the threshold voltage is.

Problem 5: Write the ideal diode equation and explain each quantity in the equation.

- A silicon diode is connected in a circuit with a 5V power supply and a series resistor of 1kΩ. Draw the circuit. The saturation current of the diode is $I_s = 10^{-12}A$. Use the diode equation to calculate the current through the diode when the voltage across it is 0.7V.

$$n = 1$$

$$\frac{kt}{q} = 26mV$$

- Find the voltage across the diode when the current through it is 1mA.

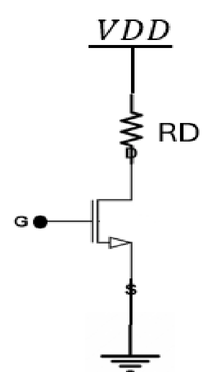
Problem 6: What is a MOSFET? What is a p-channel and n-channel MOSFET? Draw their side view and label each segment.

- A n-channel MOSFET is used in a circuit with a drain resistor RD=1kΩ and a supply voltage VDD=10V. The MOSFET has the following parameters:

V_{th}=2V

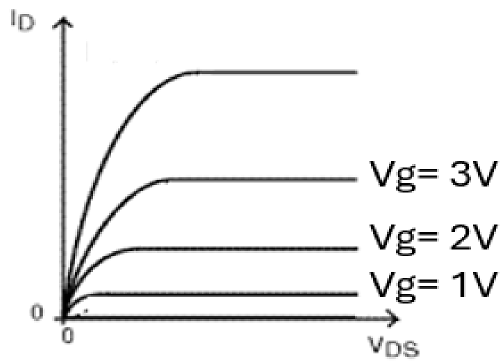
$\frac{\mu_n.C_{ox}.Z}{2L}$=0.5 $\frac{mA}{V^2}$

If VGS=4V, assuming the MOSFET is in the saturation region, calculate the current and voltage across the resistor.



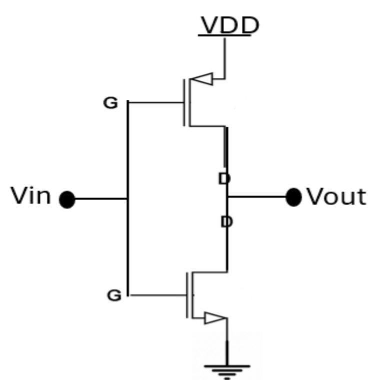
Problem 7: Draw the circuit symbol of p-channel and n-channel MOSFET.

Problem 8:



In the shown plot, draw another curve when Vg<Vth. What is the state of the N-channel MOSFET when Vg<Vth?

Problem 9: Fill up the table.



Vin	Vout
0	
VDD	

- What is this electronic device? Where can you use such a device?